
Subject: Mathematics –II (303191151)

Semester: 2nd Sem. B.Tech Programme (All Branches)

Lecture Notes: Unit – 5, Vector Calculus

Scalar point function:

- If corresponding to each point $P(x, y, z)$ of a region R in space there corresponds a unique scalar function $f = f(x, y, z)$ then, f is called a scalar point function and R is called a scalar field.
- For example,
 - (i) the temperature field in a body.
 - (ii) The pressure field of the air in the earth's atmosphere.
 - (iii) The density of a body.These quantities take different values at different points.

Note: A scalar field which is independent of time is called a stationary or steady-state scalar field.

Vector point function:

If corresponding to each point $P(x, y, z)$ of a region R in space there corresponds a unique vector function $v(x, y, z) = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$ then, v is called a vector point function and R is called a vector field.

For example,

- (i) the velocity of a moving fluid at any instant.
- (ii) The gravitational force.
- (iii) The electric and magnetic field intensity.

Note: A vector field which is independent of time is called a stationary or steady-state vector field.

Vector differential operator -

The vector differential operator is denoted by ∇ (del or nabla) and is defined as

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

Gradient of a scalar field: - For a given scalar function $\phi(x, y, z)$ the gradient of ϕ is denoted by $\text{grad } \phi$ or $\nabla\phi$ is defined as

$$\nabla\phi = \hat{i} \frac{\partial\phi}{\partial x} + \hat{j} \frac{\partial\phi}{\partial y} + \hat{k} \frac{\partial\phi}{\partial z}$$

Example: Find the gradient of $\phi = 3x^2y - y^3z^2$ at the point $(1, -2, 1)$.

Solution: We have,

$$\begin{aligned}\nabla\phi &= \hat{i} \frac{\partial\phi}{\partial x} + \hat{j} \frac{\partial\phi}{\partial y} + \hat{k} \frac{\partial\phi}{\partial z} \\ &= \hat{i}(6xy) + \hat{j}(3x^2 - 3y^2z^2) + \hat{k}(-2y^3z)\end{aligned}$$

At the point $(1, -2, 1)$

$$\nabla\phi = -12\hat{i} - 9\hat{j} - 16\hat{k}.$$

Example: Evaluate $\nabla e^{(r^2)}$, where $r^2 = x^2 + y^2 + z^2$

Solution:

$$r^2 = x^2 + y^2 + z^2$$

Differentiating r partially with respect to x, y, z

$$2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$2r \frac{\partial r}{\partial y} = 2y \Rightarrow \frac{\partial r}{\partial y} = \frac{y}{r}$$

$$2r \frac{\partial r}{\partial z} = 2z \Rightarrow \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\nabla e^{r^2} = \hat{i} \frac{\partial e^{r^2}}{\partial x} + \hat{j} \frac{\partial e^{r^2}}{\partial y} + \hat{k} \frac{\partial e^{r^2}}{\partial z} = \hat{i} \frac{\partial e^{r^2}}{\partial r} \frac{\partial r}{\partial x} + \hat{j} \frac{\partial e^{r^2}}{\partial r} \frac{\partial r}{\partial y} + \hat{k} \frac{\partial e^{r^2}}{\partial r} \frac{\partial r}{\partial z}$$

$$= \hat{i}(2r e^{r^2}) \frac{x}{r} + \hat{j}(2r e^{r^2}) \frac{y}{r} + \hat{k} 2r e^{r^2} \frac{z}{r} = 2e^{r^2}(x\hat{i} + y\hat{j} + z\hat{k})$$

Example: Find a unit normal vector to the surface $x^3 + y^3 + 3xyz = 3$ at the point $(1, 2, -1)$

Solution:

$$\phi(x, y, z) = x^3 + y^3 + 3xyz - 3 = 0$$

$$\nabla\phi = \hat{i} \frac{\partial\phi}{\partial x} + \hat{j} \frac{\partial\phi}{\partial y} + \hat{k} \frac{\partial\phi}{\partial z}$$

$$= \hat{i} (3x^2 + 3yz) + \hat{j} (3y^2 + 3xz) + \hat{k} (3xy)$$

At the point $(1, 2, -1)$

$$\nabla\phi = -3\hat{i} + 9\hat{j} + 6\hat{k}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{-3\hat{i} + 9\hat{j} + 6\hat{k}}{\sqrt{126}}$$

Examples for Practice:

1. Find a unit normal vector to the surface $x^2y + 3xz^2 = 8$ at the point $(1, 0, 2)$
2. Find the unit normal to the surface $x^2 + xy + y^2 + xyz$ at the point $(1, -2, 1)$

Directional Derivative: -

The directional derivative of scalar point function $\phi(x, y, z)$ in the direction of vector $\hat{a}$, is the component of $\nabla\phi$ in the direction of $\hat{a}$.

If $\hat{a}$ is the unit vector in the direction of a , then the directional derivative of ϕ in the direction of a is $D\phi = \nabla\phi \cdot \hat{a}$

Examples: Find the directional derivative of $\phi(x, y, z) = x^3 - xy^2 - z$ at point $(1, 1, 0)$ in the direction of $v = 2\hat{i} - 3\hat{j} + 6\hat{k}$

Sol. Here, $\phi(x, y, z) = x^3 - xy^2 - z$

$$\text{Solution: } \text{grad } \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} = \hat{i} (6x^2y + 24y^2z - 8z^2x) + \hat{j} (6x^2 - 2y) + \hat{k} (-1)$$

At the point $(1, 1, 0)$

$$\nabla\phi = 2\hat{i} - 2\hat{j} - \hat{k}$$

The direction derivative of ϕ at point $P(1, 1, 0)$ in the direction of v is

$$D\phi = \nabla\phi \cdot \hat{v} = \nabla\phi \cdot \frac{v}{|v|} = (2\hat{i} - 2\hat{j} - \hat{k}) \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) / \sqrt{49} = \frac{4}{7}$$

Example: Find the directional derivative of $\phi = 6x^2y + 24y^2z - 8z^2x$ at $(1, 1, 1)$ in the direction of $v = 2\hat{i} - 2\hat{j} + \hat{k}$. Hence, find the maximum value.

Solution:

$$\nabla\phi = (12xy - 8z^2)\hat{i} + (6x^2 + 48yz)\hat{j} + (24y^2 - 16zx)\hat{k}$$

□

$$\phi_{(1,1,1)} = 4\hat{i} + 54\hat{j} + 8\hat{k}$$

Directional derivative in the direction of $v = (2i - 2j + k)$ at the point $(1,1,1)$

$$\begin{aligned}
 &= \nabla \phi \frac{v}{|v|} \\
 &= (4i + 54j + 8k) \frac{2i - 2j + k}{|2i - 2j + k|} \\
 &= (4i + 54j + 8k) \frac{(2i - 2j + k)}{\sqrt{4 + 4 + 1}} \\
 &= \frac{8 - 108 + 8}{3} = -\frac{92}{3}
 \end{aligned}$$

Maximum value of directional derivative = $|\nabla \phi|$

Example for Practice

1. Find the directional derivative of $\phi(x, y, z) = xy^2 + yz^3$ at the point $P(2, -1, 1)$ in the direction of PQ where Q is the point $(3, 1, 3)$
2. In what direction from $(-1, 1, 2)$ is the directional derivative of $\phi = xy^2z^3$ a maximum? Find also the magnitude of this maximum.
3. Find the directional derivative of the scalar function $\phi = xyz$ in the direction of the outer normal to the surface $z = xy$ at the point $(3, 1, 3)$.
4. Find the directional derivative of $\phi = xy + yz + zx$ at $(1, 2, 0)$ in the direction of $\hat{i} + 2\hat{j} + 2\hat{k}$

Divergence of a vector function

Let $F = F_1 i + F_2 j + F_3 k$ be a vector function then, divergence of F is

$$\begin{aligned} \text{div } F = \nabla F &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (F_1 i + F_2 j + F_3 k) \\ &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \end{aligned}$$

Note:

1. If $\nabla F = 0$ then, the vector function F is called solenoidal or incompressible
2. In hydrodynamics (the study of fluid motion), a velocity field that is divergence free is called **incompressible**.
3. In the study of electricity and magnetism, a vector field that is divergence free is called **solenoidal**.

Example: If $F = x^2 z i - 2y^3 z^3 j + xy^2 z k$ then, find divergence of F at $(1, -1, 1)$

Sol. Here, $F = x^2 z i - 2y^3 z^3 j + xy^2 z k$

$$\begin{aligned} \nabla F &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \\ &= \frac{\partial(x^2 z)}{\partial x} + \frac{\partial(-2y^3 z^3)}{\partial y} + \frac{\partial(xy^2 z)}{\partial z} \\ &= 2xz - 6y^2 z^3 + xy^2 \end{aligned}$$

$$\nabla F_{(1,-1,1)} = -3$$

Example: Show that $A = 3y^4 z^2 i + 4x^3 z^2 j - 3x^2 y^2 k$ is a solenoidal. **Solution:**

$$\begin{aligned} \nabla \bar{A} &= \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \\ &= \frac{\partial(3y^4 z^2)}{\partial x} + \frac{\partial(4x^3 z^2)}{\partial y} + \frac{\partial(-3x^2 y^2)}{\partial z} = 0 \end{aligned}$$

Hence Given function is solenoidal.

Example for Practice:

1. Determine the constant a such that $\mathbf{A} = (ax^2y + yz)\hat{i} + (xy^2 + xz^2)\hat{j} + (2xyz - 2x^2y^2)\hat{k}$ is solenoidal.
2. Find $\text{div } \mathbf{F}$, where $\mathbf{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$
3. If $\vec{F} = xy^2\hat{i} + 2x^2yz\hat{j} - 3yz^2\hat{k}$ find $\nabla \cdot \mathbf{F}$ at a point (1,-1,1)
4. If $\vec{F} = (x^2 - y^2 + 2xz)\hat{i} + (xz - xy + yz)\hat{j} - (z^2 + x^2)\hat{k}$ then find $\nabla \cdot \mathbf{F}$.

Curl

Let $\mathbf{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ be a vector function then, curl of $\mathbf{F}$ is

$$\text{curl } \mathbf{F} \text{ or } \nabla \times \mathbf{F} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{bmatrix}$$

Note: - If $\nabla \times \mathbf{F} = 0$ then, the vector function $\mathbf{F}$ is called Irrotational or conservative.

Example: If $\mathbf{F} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$ then, find curl of $\mathbf{F}$ at (1,-1,1)

Solution: Here, $\mathbf{F} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$

$$\nabla \times \mathbf{F} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{bmatrix}$$

$$= \hat{i}(2z^4 + 2x^2y) + \hat{j}(3xz^2) + \hat{k}(-4xyz)$$

At point (1, -1, 1)

$$\nabla \times \mathbf{F} = 3\hat{j} + 4\hat{k}$$

Example: Show that $r = x\hat{i} + y\hat{j} + z\hat{k}$ is Irrotational. Solution:

$$\nabla \times F = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{bmatrix}$$

$$= 0\hat{i} + 0\hat{j} + 0\hat{k}$$

Therefore, $r = x\hat{i} + y\hat{j} + z\hat{k}$ is Irrotational.

Example: Find curl of $A = e^{xyz}(i + j + k)$ at the point (1,2,3)

Solution:

$$\nabla \times A = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{xyz} & e^{xyz} & e^{xyz} \end{bmatrix}$$

$$= \hat{i}(e^{xyz}(xz) - e^{xyz}(xy)) - \hat{j}(e^{xyz}(yz) - e^{xyz}(xy)) + \hat{k}(e^{xyz}(yz) - e^{xyz}(xz))$$

$$= e^{xyz}(\hat{i}(xz - xy) - \hat{j}(yz - xy) + \hat{k}(yz - xz))$$

$$\nabla \times A_{(1,2,3)} = e^6(i(3 - 2) - j(6 - 2) + k(6 - 3)) = e^6(i - 4j + 3k)$$

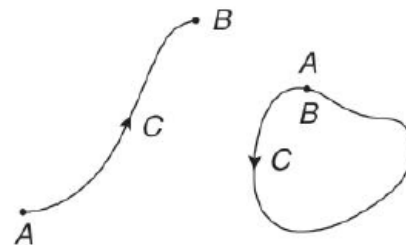
Example for Practice:

1. If $\vec{F} = xy^2\hat{i} + 2x^2yz\hat{j} - 3yz^2\hat{k}$ find $\nabla \times F$ at a point (1,-1,1)
2. If $\vec{F} = (x^2 - y^2 + 2xz)\hat{i} + (xz - xy + yz)\hat{j} - (z^2 + x^2)\hat{k}$ then find $\nabla \times F$.
3. Find **div(grad ϕ)** and **curl(grad ϕ)** at (1,1,1) for $\phi = x^2y^3z^4$

LINE INTEGRAL:

The line integral is a simple generalization of a definite integral $\int_a^b f(x) dx$ which is integrated from $x = a$ (point A) to $x = b$ (point B) along the x -axis.

In a line integral the integration is done along a curve C in space.



Let $\vec{F}(\vec{r})$ be a vector function defined at every point of a curve C . If $(\vec{r})$ is the position vector of the point $P(x, y, z)$ on the curve C then the line integral of $\vec{F}(\vec{r})$ over a curve C is defined by

$$\int_C \vec{F}(\vec{r}) d\vec{r} = \int_C (F_1 dx + F_2 dy + F_3 dz)$$

where $\vec{F} = F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}$ & $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

If the curve C is represented by parametric representation, $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$

Then the line integral along the curve C from $t = a$ to $t = b$ is

$$\int_C \vec{F}(\vec{r}) d\vec{r}$$

$$= \int_a^b \vec{F} \frac{d\vec{r}}{dt} dt = \int_a^b \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt$$

If C is closed curve, then the symbol of the line $\int_C$ integral is replaced $\oint_C$

Example

If $\vec{F} = 3xy\hat{i} - y^2\hat{j}$, $\int_C \vec{F}(\vec{r})d\vec{r}$ evaluate where C is the curve $y = 2x^2$ from (0,0) to (1,2).

Solution:

Given $\vec{F} = 3xy\hat{i} - y^2\hat{j}$, $d\vec{r} = dx\hat{i} + dy\hat{j} \Rightarrow \vec{F} \cdot d\vec{r} = 3xy - y^2$

Given C is $y = 2x^2$,

$$\therefore dy = 4x dx$$

Along C, x varies from 0 to 1

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_0^1 3x(2x^2) dx - 4x^4(4x dx) \\ &= \int_0^1 (6x^3 - 16x^5) dx = \left[6\frac{x^4}{4} - 16\frac{x^6}{6} \right]_0^1 = \frac{6}{4} - \frac{16}{6} = -\frac{7}{6} \text{ units.} \end{aligned}$$

Example

Find the work done when a force $\vec{F} = (x^2 - y^2 + x)\hat{i} - (2xy + y)\hat{j}$, moves a particle from the origin to the point (1, 1) along $y^2 = x$.

Solution:

Given $\vec{F} = (x^2 - y^2 + x)\hat{i} - (2xy + y)\hat{j}$,

$$d\vec{r} = dx\hat{i} + dy\hat{j} \Rightarrow \vec{F} \cdot d\vec{r} = (x^2 - y^2 + x)dx - (2xy + y)dy$$

Given C is $y^2 = x$, $\therefore 2ydy = dx$

Along C, x varies from 0 to 1

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_0^1 ((y^2)^2 - y^2 + y^2)2ydy - (2y^3 + y)dy \\ &= \int_0^1 (2y^5 - 2y^3 + 2y^3 - 2y^3 - y) dy \\ &= \int_0^1 (2y^5 - 2y^3 - y) dy = \left[\frac{2y^6}{6} - \frac{2y^4}{4} - \frac{y^2}{2} \right]_0^1 = \frac{2}{6} - \frac{2}{4} - \frac{1}{2} = -\frac{2}{3} \end{aligned}$$

Examples

Find the work done in moving a particle in the force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} -$

$z\hat{k}$ from $t=0$ to 1 along the curve $x = 2t^2, y = t, z = 4t^3$.

Examples

Find $\int_C \vec{F} \cdot d\vec{r}$ where C is the circle $x^2 + y^2 = 4$ in the xy-plane where

$$\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} - 3xz^2\hat{k}.$$

SURFACE INTEGRAL:

An integral which is evaluated over a surface is called a surface integral. Consider a surface S. Let $\vec{F}$ be a vector valued function which is defined at each point on the surface and let P be any point on the surface and $\vec{n}$ be the unit outward normal to the surface at P. The normal component of $\vec{F}$ at P is $\vec{F} \cdot \vec{n}$.

The integral of the normal component of $\vec{F}$ is denoted by $\iint_S \vec{F} \cdot \vec{n} \, ds$

EVALUATION OF SURFACE INTEGRAL

If R_1 be the projection of S on the xy -plane, $\hat{k}$ is the unit vector normal the xy -plane then

$$ds = \frac{dx dy}{|\bar{n} \hat{k}|}$$

$$\therefore \iint_S \bar{F} \cdot \bar{n} ds = \iint_{R_1} \bar{F} \cdot \bar{n} \frac{dx dy}{|\bar{n} \hat{k}|}$$

If R_2 be the projection of S on the yz -plane, $\hat{i}$ is the unit vector normal the yz -plane then

$$ds = \frac{dy dz}{|\bar{n} \hat{i}|}$$

$$\therefore \iint_S \bar{F} \cdot \bar{n} ds = \iint_{R_2} \bar{F} \cdot \bar{n} \frac{dy dz}{|\bar{n} \hat{i}|}$$

If R_3 be the projection of S on the xz -plane, $\hat{j}$ is the unit vector normal the xz -plane then

$$ds = \frac{dx dz}{|\bar{n} \hat{j}|}$$

$$\therefore \iint_S \bar{F} \cdot \bar{n} ds = \iint_{R_3} \bar{F} \cdot \bar{n} \frac{dx dz}{|\bar{n} \hat{j}|}$$

Problems based on Surface Integral

Example

Evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ if $\vec{F} = (x + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x + y + 2z = 6$ in the first octant.

Solution:

$$\text{Given } \vec{F} = (x + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$$

$$\text{Let } \varphi = 2x + y + 2z + 6$$

$$\text{then } \nabla\varphi = \hat{i} \frac{\partial\varphi}{\partial x} + \hat{j} \frac{\partial\varphi}{\partial y} + \hat{k} \frac{\partial\varphi}{\partial z} = 2\hat{i} + 1\hat{j} + 2\hat{k}$$

$$|\nabla\varphi| = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

$$\hat{n} = \frac{\nabla\varphi}{|\nabla\varphi|} = \frac{2\hat{i} + 1\hat{j} + 2\hat{k}}{3}$$

$$\vec{F} \cdot \hat{n} = [(x + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}] \cdot \left(\frac{2\hat{i} + 1\hat{j} + 2\hat{k}}{3} \right)$$

$$= \frac{1}{3} [2(x + y^2) - 2x + 4yz]$$

$$= \frac{2}{3} [y^2 + 2yz]$$

$$= \frac{2}{3} y[y + 2z]$$

$$= \frac{2}{3} [y + 6 - 2x - y] \quad [\because 2z = 6 - 2x - y]$$

$$= \frac{2}{3} [6 - 2x]$$

$$= \frac{4}{3} y[3 - x]$$

Let R be the projection of S on the xy – plane

$$\therefore ds = \frac{dx dy}{|\bar{n} \hat{k}|}$$

$$\bar{n} \cdot \hat{k} = \left(\frac{2\hat{i} + 1\hat{j} + 2\hat{k}}{3} \right) \cdot \hat{k} = \frac{2}{3}$$

$$\therefore \iint_S \bar{F} \cdot \bar{n} ds = \iint_R \bar{F} \cdot \bar{n} \frac{dx dy}{|\bar{n} \hat{k}|} = \iint_R \frac{4}{3} y [3 - x] \cdot \frac{dx dy}{\left(\frac{2}{3}\right)}$$

$$= 2 \iint [3 - x] y dx dy$$

In R_1 ($2x + y = 6$), x varies from 0 to $\frac{6-y}{2}$

y varies from 0 to 6

$$= 2 \int_0^6 \int_0^{\frac{6-y}{2}} y(3 - x) dx dy$$

$$= 2 \int_0^6 \left[3x - \frac{x^2}{2} \right]_0^{\frac{6-y}{2}} dy$$

$$= 2 \int_0^6 \frac{1}{2} (18y - 3y^2) - \frac{1}{8} (6 - y^2) dy$$

$$= \frac{2}{2} \left[\frac{18y^2}{2} - \frac{3y^3}{3} - \frac{(6 - y)^3}{8(3)(-1)} \right]$$

$$= \left[9(6)^2 - (6)^3 + \frac{1}{12} (0) \right] - \left[0 - 0 + \frac{1}{12} (6^2) \right]$$

$$= 81 \text{ units.}$$

Example

Evaluate $\iint_S 6xy \, ds$ where S is the portion of the plane $x + y + z = 1$ that lies in front of yz plane.

Solution:

We are looking for portion of the plane ABC that lies in front of the yz – plane, therefore, we write equation of the surface in the form $x = f(y, z)$

For the points on the surface we have $x = 1 - y - z$

$$\begin{aligned}\iint_S 6xy \, ds &= \iint_S 6(1 - y - z)y\sqrt{3} \, dA \\ &= 6\sqrt{3} \int_0^1 \int_0^{1-y} 6(1 - y - z)y \, dz \, dy \\ &= 6\sqrt{3} \int_0^1 \left[yz - y^2z - \frac{1}{2}yz^2 \right]_0^{1-y} dy \\ &= 6\sqrt{3} \left[\frac{1}{4}y^2 - \frac{1}{3}y^3 + \frac{1}{8}y^4 \right]_0^1 = \frac{\sqrt{3}}{4}\end{aligned}$$

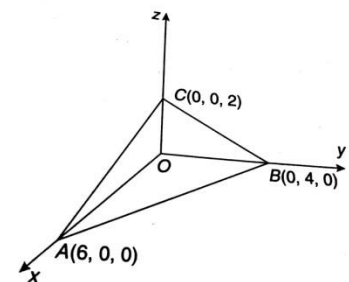
Example

Evaluate $\iint_S \vec{F} \cdot \hat{n} \, dS$, where $\vec{F} = 18z\hat{i} - 12\hat{j} + 3y\hat{k}$ and S is the part of the plane $2x + 3y + 6z = 12$ in the first octant.

Solution: The given surface is the plane $2x + 3y + 6z = 12$ in the first octant.

Let $\phi = 2x + 3y + 6z$

$$\begin{aligned}\hat{n} &= \frac{\nabla\phi}{|\nabla\phi|} \\ &= \frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{\sqrt{4 + 9 + 36}} \\ &= \frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{7}\end{aligned}$$



Let R be the projection of the plane $2x + 3y + 6z = 12$ on the xy -plane, which is a triangle OAB bounded by the lines $y = 0$, $x = 0$ and $2x + 3y = 12$

$$dS = \frac{dx dy}{|\hat{n} \cdot \hat{k}|} = \frac{7}{6} dx dy$$

Along the vertical strip PQ, y varies from 0 to $\frac{12-2x}{3}$ and in the region R, x varies from 0 to 6

$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} dS &= \iint_R (18z\hat{i} - 12\hat{j} + 3y\hat{k}) \cdot \left(\frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{7} \right) \frac{7}{6} dx dy \\ &= \frac{1}{6} \iint_R (36z - 36 + 18y) dx dy \\ &= 3 \iint_R \left[2 \left(\frac{12-2x-3y}{6} \right) - 2 + y \right] dx dy \\ &= \int_0^6 \int_0^{\frac{12-2x}{3}} (6 - 2x) dy dx \\ &= 2 \int_0^6 (3 - x) \left| y \right|_0^{\frac{12-2x}{3}} dx \\ &= 2 \int_0^6 (3 - x) \left(\frac{12-2x}{3} \right) dx \\ &= \frac{4}{3} \int_0^6 (x^2 - 9x + 18) dx = 24 \end{aligned}$$

GREEN'S THEOREM

Statement: If $M(x, y), N(x, y), \frac{\partial M}{\partial y}, \frac{\partial N}{\partial x}$ be continuous everywhere in a region R of xy plane

bounded by a closed curve c, then $\oint_c M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

Example

Verify Green's Theorem for $\oint_C [(x^2 - 2xy)dx + (x^2y + 3)dy]$ where C is the boundary of the region bounded by the parabola $y = x^2$ and the line $y = x$.

Solution: The points of intersection of the parabola $y = x^2$ and the line $y = x$ are obtained as $x = x^2, x = 0, 1$ and $y = 0, 1$

Hence, $O(0,0)$ and $B(1,1)$ are the points of intersection.

$$M = x^2 - 2xy, N = x^2y + 3$$

$$\frac{\partial M}{\partial y} = -2x, \quad \frac{\partial N}{\partial x} = 2xy$$

$$\oint_C (Mdx + Ndy) = \int_{OAB} (Mdx + Ndy) + \int_{BO} (Mdx + Ndy)$$

Along OAB: $y = x^2, dy = 2xdx$
x varies from 0 to 1

$$\begin{aligned} \int_{OAB} (Mdx + Ndy) &= \int_{OAB} [(x^2 - 2xy)dx + (x^2y + 3)dy] \\ &= \int_0^1 [x^2 - 2x \cdot x^2]dx + (x^2 \cdot x^2 + 3)2x dx \\ &= \int_0^1 (x^2 - 2x^3 + 2x^5 + 6x)dx = \frac{19}{6} \end{aligned}$$

Along BO: $y = x, dy = dx$
x varies from $x = 1$ to 0

$$\begin{aligned} \int_{BO} (Mdx + Ndy) &= \int_{BO} [(x^2 - 2xy)dx + (x^2y + 3)dy] \\ &= \int_1^0 [(x^2 - 2x^2)dx + (x^3 + 3)dx] = -\frac{35}{12} \end{aligned}$$

$$\text{Substituting in (1), } \oint_C (Mdx + Ndy) = \frac{19}{6} - \frac{35}{12} = \frac{1}{4} \quad \text{----- (2)}$$

Let R be the region bounded by the line $y = x$ and the parabola $y = x^2$

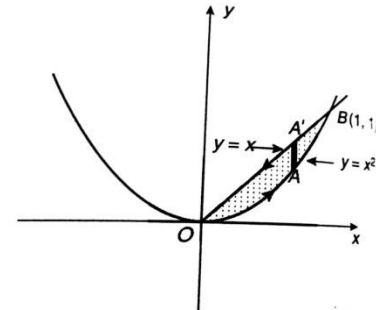
Along the vertical strip AA', y varies from x^2 to x and in the region R, x varies from 0 to 1

$$\begin{aligned} \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \int_0^1 \int_{x^2}^x (2xy + 2x) dy dx \\ &= \int_0^1 [xy^2 + 2xy]_{x^2}^x dx \\ &= \int_0^1 (x^3 + 2x^2 - x^5 - 2x^3) dx \\ &= \frac{1}{4} \quad \text{----- (3)} \end{aligned}$$

From equations (2) and (3),

$$\oint_C (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \frac{1}{4}$$

Hence, Green's theorem is verified.



----- (1)

Example

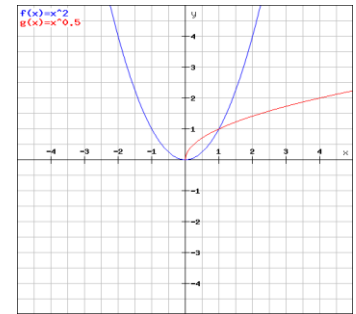
Using Green's Theorem, evaluate $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region bounded by $y^2 = x$ and $y = x^2$.

Solution: $y^2 = x$ and $y = x^2$ are two parabolas intersecting at $(0,0)$ and $(1,1)$

Here, $M = 3x^2 - 8y^2$, $N = 4y - 6xy$

$$\frac{\partial M}{\partial y} = -16y, \quad \frac{\partial N}{\partial x} = -6y$$

$$\begin{aligned} \oint_C (Mdx + Ndy) &= \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \\ &= \int_0^1 \int_{x^2}^{\sqrt{x}} 10y \, dx \, dy \\ &= \int_0^1 5(y^2)_{x^2}^{\sqrt{x}} dx \\ &= 5 \int_0^1 (x - x^4) dx = \frac{3}{2} \end{aligned}$$



Example

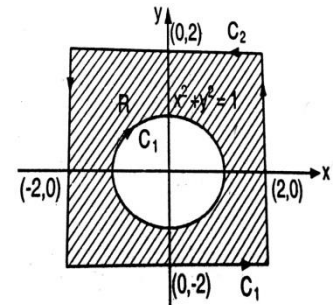
Evaluate $\oint_C \left[-\frac{y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy \right]$ where $C = C_1 \cup C_2$, with $C_1: x^2 + y^2 = 1$ and $C_2: x = \pm 2, y = \pm 2$.

Solution: Here, $M = -\frac{y}{x^2+y^2}$, $N = \frac{x}{x^2+y^2}$

$$\frac{\partial M}{\partial x} = \frac{y^2 - x^2}{(x^2+y^2)^2}, \quad \frac{\partial N}{\partial y} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

Which are continuous on the region R bounded by C .

$$\begin{aligned} \oint_C \left[-\frac{y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy \right] &= \iint_R \left[\frac{y^2 - x^2}{(x^2+y^2)^2} - \frac{y^2 - x^2}{(x^2+y^2)^2} \right] dx dy \\ &= 0 \end{aligned}$$



Example

State Green's Theorem and use it to find the work done by $F = (4x - 2y)i + (2x - 4y)j$ in moving a particle once clockwise around the circle $(x - 2)^2 + (y - 2)^2 = 4$

Solution: $F \cdot dr = (4x - 2y)dx + (2x - 4y)dy$
 $= Mdx + Ndy$

$$M = 4x - 2y, \quad N = 2x - 4y$$

$$\frac{\partial M}{\partial y} = -2, \quad \frac{\partial N}{\partial x} = 2$$

By Green's Theorem, $\oint_C F \cdot dr = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$

$$= \iint_R (2 + 2) dx dy = 4 \iint_R dx dy \\ = 4 (4\pi) = 16\pi$$

DIVERGENCE THEOREM: (Convert surface integral to volume integral)

Statement: If $\vec{F}$ be a vector point function having continuous partial derivatives in the region bounded by a closed surface S , then

$$\iint_S \vec{F} \cdot \vec{n} ds = \iiint_V \text{div} \vec{F} dv$$

Example

Find the flux of $F = yzj + z^2k$ outward through the surface S cut from the cylinder $y^2 + z^2 = 1, z \geq 0$ by the plane $x=0$ & $x=1$

Solution: The outward normal field on S calculated from the gradient of $g(x, y, z) = y^2 + z^2$ to be $\vec{n} = \frac{\nabla g}{|\nabla g|} = \frac{2y j + 2z k}{\sqrt{4y^2 + 4z^2}} = \frac{2y j + 2z k}{2\sqrt{1}} = yj + zk$

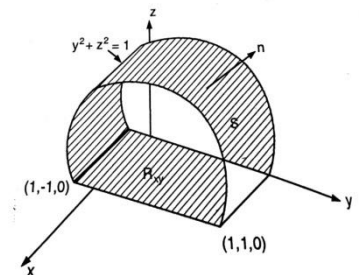
$$dS = \frac{|\nabla g|}{|\nabla g \cdot k|} dA = \frac{2}{|2z|} dA = \frac{1}{z} dA$$

Since $z \geq 0$ on S

$$\begin{aligned} \vec{F} \cdot \vec{n} &= (yzj + z^2k) \cdot (yj + zk) \\ &= y^2z + z^3 \\ &= z(y^2 + z^2) = z \end{aligned}$$

Therefore, the flux F outward through S is

$$\iint_S \vec{F} \cdot \vec{n} dS = \iint_S z \left(\frac{1}{z} dA \right) = \iint_{R_{xy}} dA = \text{area } R_{xy} = 2$$



Example

Find the flux of $F = 4xzi - y^2j + yzk$ outward through the surface of the cube cut from the first octant by the planes $x = 1, y = 1, z = 1$

Solution: Here $F = 4xzi - y^2j + yzk$

$$\nabla \cdot F = \frac{\partial}{\partial x} (4xz) + \frac{\partial}{\partial y} (-y^2) + \frac{\partial}{\partial z} (yz) = 4z - 2y + y$$

$$\therefore \nabla \cdot F = 4z - y$$

Over the interior of cube:

$$\text{Flux} = \iiint \vec{F} \cdot \vec{n} ds = \iiint \nabla \cdot F dV = \int_0^1 \int_0^1 \int_0^1 (4z - y) dx dy dz$$

VOLUME INTEGRAL:

$$\iiint_V \phi dv = \iiint_V \phi(x, y, z) dx dy dz = \iiint_V F dv$$

Example

If $\phi = 45x^2y$ then evaluate $\iiint_V \phi dv$ where V denote the closed region bounded by the planes $4x + 2y + z = 8, x = 0, y = 0, z = 0$.

$$\begin{aligned} \text{Solution: } \iiint \phi dV &= \int_{x=0}^2 \int_{y=0}^{4-2x} \int_{z=0}^{8-4x-2y} 45 x^2 y dx dy dz \\ &= 45 \int_{x=0}^2 \int_{y=0}^{4-2x} x^2 y (8 - 4x - 2y) dy dx \\ &= 45 \int_{x=0}^2 \frac{1}{3} x^2 (4 - 2x)^3 dx = 128 \end{aligned}$$

STOKE'S THEOREM:

If s is an open two sided surface bounded by a closed non intersecting curve and if a vector function $F(x, y, z)$ has continuous first partial derivatives in a domain in a space containing s Then

$$\oint_C F \cdot dr = \iint_S (\text{curl} F) \cdot n ds = \iint_S (\nabla \times F) \cdot n ds$$

Where C is described in positive (anti clock wise) direction and $\hat{n}$ is a unit positive(outward drawn) normal to s .

Example

Verify Stoke's theorem for $A = (2x - y)i - yz^2j - y^2zk$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.

Solution: The boundary C of S is a circle in the xy -plane of radius unity and centre at the origin. Let $x = \cos t, y = \sin t, z = 0, 0 \leq t \leq 2\pi$ be the parametric equations of C

$$\begin{aligned} \text{Then, } \oint_C A \cdot dr &= \oint_C [(2x - y)dx - yz^2 dy - y^2 z dz] \\ &= \int_0^{2\pi} (2 \cos t - \sin t)(-\sin t) dt \\ &= \int_0^{2\pi} (-2 \sin t \cos t + \sin^2 t) dt = \pi \end{aligned}$$

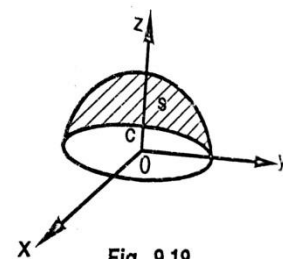


Fig. 0.10

$$\text{Also, } \nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix} = \mathbf{k}$$

$$\text{curl } \mathbf{A} \cdot \mathbf{n} = \mathbf{k} \cdot \mathbf{k} = 1$$

$\iint_S (\text{curl } \mathbf{A}) \cdot \mathbf{n} \, dS = \iint_R dx \, dy$, where R is the projection of S on the xy-plane

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \, dx$$

$$= \int_{-1}^1 2\sqrt{1-x^2} \, dx$$

$$= 4 \int_0^1 \sqrt{1-x^2} \, dx = \pi$$

Hence, Stoke's Theorem is verified.

Example

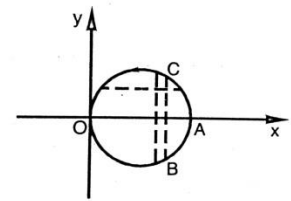
Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ taken over the portion of the surface $x^2 + y^2 - 2ax + az = 0$ and the bounding curve in the plane $z=0$ and $\mathbf{F} = (y^2 + z^2 - x^2)\mathbf{i} + (z^2 + x^2 - y^2)\mathbf{j} + (x^2 + y^2 - z^2)\mathbf{k}$

Solution: The given surface meets the plane $z = 0$ in the circle $x^2 + y^2 - 2ax = 0, z = 0$

$$\mathbf{F} = (y^2 + z^2 - x^2)\mathbf{i} + (z^2 + x^2 - y^2)\mathbf{j} + (x^2 + y^2 - z^2)\mathbf{k}$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 + z^2 - x^2 & z^2 + x^2 - y^2 & x^2 + y^2 - z^2 \end{vmatrix}$$

$$= i(2y - 2z) + j(2z - 2x) + k(2x - 2y)$$



The surface integral of $\nabla \times \mathbf{F}$ over the given surface is the same as the surface integral of $\nabla \times \mathbf{F}$ over the area of the circle $x^2 + y^2 - 2ax = 0, z = 0$

$$dS = n \, dS = k \, dx \, dy$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_0^{2a} \int_{-\sqrt{2ax-x^2}}^{\sqrt{2ax-x^2}} (2x - 2y) \, dy \, dx$$

$$\begin{aligned} &= \int_0^{2a} \int_{-\sqrt{2ax-x^2}}^{\sqrt{2ax-x^2}} 2x dy dx - \int_0^{2a} \int_{-\sqrt{2ax-x^2}}^{\sqrt{2ax-x^2}} 2y dy dx \\ &= 2 \int_0^{2a} \int_0^{\sqrt{2ax-x^2}} 2x dy dx \\ &= 4 \int_0^{2a} x [y]_0^{\sqrt{2ax-x^2}} = 2\pi a^2 \end{aligned}$$